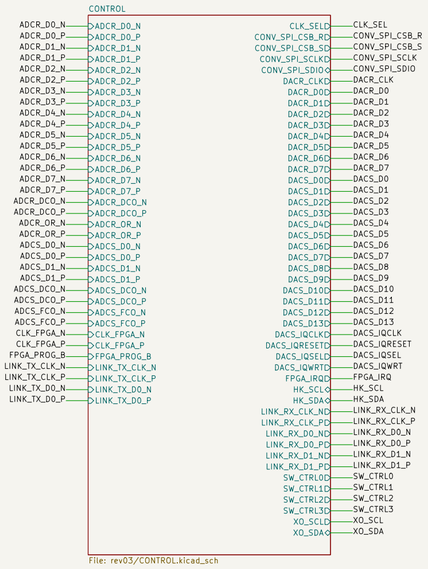
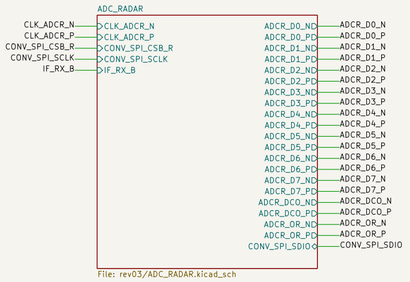
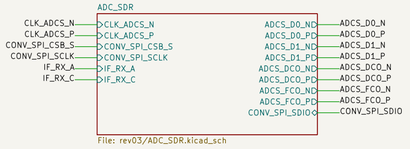
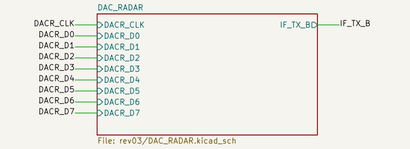


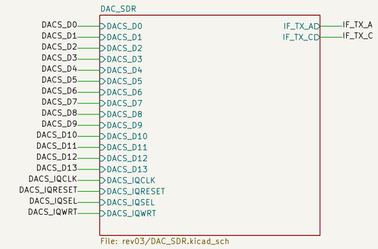














Derived from the Allied Vision CSI-2 adapter for Kria KV260 (MIT), see README

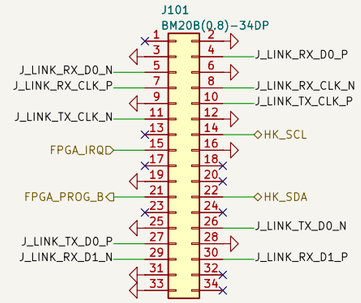
KV260-IAS2RF

Sheet: /
File: CSI2-KRIA-KV260_Rev01.kicad_sch

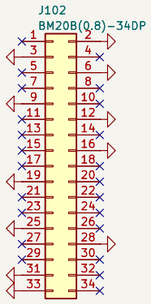
Title: KV260-IAS2RF Rev03 - root: sheet hierarchy

Size: A2	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 1/10

J101: IAS1 (KV260 J8), the link. Pins per INTERFACE_SPEC §2–4.



J102: IAS0 (KV260 J7), mechanical only: grounds, nothing else.



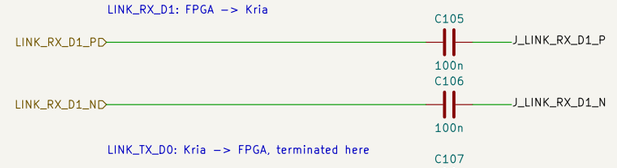
LINK_RX_D0: FPGA –> Kria



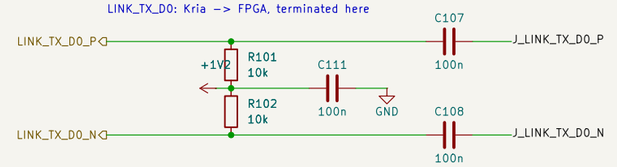
LINK_RX_CLK: FPGA –> Kria



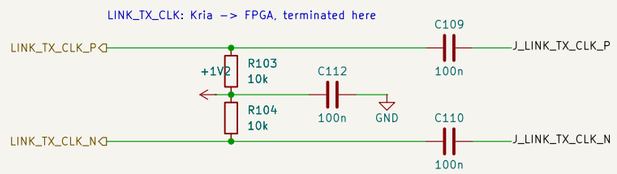
LINK_RX_D1: FPGA –> Kria



LINK_TX_D0: Kria –> FPGA, terminated here



LINK_TX_CLK: Kria –> FPGA, terminated here



IAS_LINK – how this sheet works

1. Five LVDS pairs to the Kria's HPA bank at 1.2 V: every pair AC-coupled (INTERFACE_SPEC 6.1).
2. Kria –> FPGA pairs: 2 x 49.9 ohm to +1V2 terminate and re-bias them for the FPGA.
3. FPGA –> Kria pairs: terminated and biased inside the SOM (DIFF_TERM_ADV, DQS_BIAS).
4. Pins 26/27 and 29/30 are N before P on the connector and on the carrier alike (3.2).
5. HK_SCL/SDA, FPGA_PROG_B, FPGA_IRQ are the carrier's 1.8 V levels: CONTROL translates.

Generated by _work/sheets/ias_link.py – review before editing by hand

KV260-IAS2RF

Sheet: /IAS_LINK/

File: IAS_LINK.kicad_sch

Title: IAS_LINK – Hirose IAS1 link to the Kria, AC coupling, terminations

Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 2/10

100R across each pair at the FPGA. The banks are 3.3 V, where DIFF_TERM is not allowed (UG471)

Bank 14, VCCO 3.3 V: AD9484 on AERIS-10's balls, AD9253, link inputs, CLK_FPGA, flash

Bank 15, VCCO 3.3 V: DACs (22R series), Pmod, RF switches, clock select, XO I2C

Bank 34, VCCO 2.5 V: the link pairs the FPGA drives
(LVDS_25 outputs need 2.5 V). 14 balls unused.

Bank 35, VCCO 1.8 V: ADAR1000 control, converter SPI, HK, IRQ

CLK_FPGA: ADCLK948 610–960 mV > LVDS_25 600 mV max (DS181 Table 11).
24.9R series + 24.9R shunt per leg: 50R to the line, half the swing.

Master SPI x4, UG470 Figure 2-14: M[2:0] = 001

22R series on every DAC line, as AERIS-10

Pmod, from the KV260's x2 (INTERFACE_SPEC 4.2). Pmod pin k = pad 2k-1,
pin k+6 = pad 2k. Its 3.3 V (pins 6, 12) is not used.

JTAG: Xilinx 2x7 2 mm, DS593 Figure 14

MT25QL128: FCS_B 2.4k, W and HOLD 4.7k (UG470 Figure 2-14); CCLK 22R series as AERIS

HK I2C pull-ups at 1.8 V (the carrier adds 2.21k). HK_ALERT pull-up; FPGA_IRQ pull-down

TMP1075 board temperature, I2C 0x48, ALERT to the FPGA (bank 35, H3)

M24C02 board ID EEPROM, I2C 0x50

WC high: write-protected; close JP1101 to write

XADC: VCCADC from VCCAUX through a ferrite, GNDADC to ground through another
(UG480 Figure 1-2, on-chip reference)

CFGBVS high: banks 0, 14, 15 at 3.3 V (UG470). XADC inputs and on-chip reference to
GNDADC (UG480 Figure 1-2); temperature diode unused.

PROGRAM_B: 4.7k pull-up above; push button (UG470 Table 2-4) and the Kria's
FPGA_PROG_B (1.8 V) through a 74LVC1G07 open-drain buffer at 1.8 V.

VCCBATT to GND: no bitstream encryption (DS181; UG470 Figure 2-14)

Decoupling per UG483 Table 2-2 (FTG256, XC7A50T); types per Table 2-5

VCCINT, bottom face under the BGA (on ball vias)

LINK_TX termination at the receiver (bottom face, on the ball vias)

To AERIS-10: 4 lines per ADAR1000 (CS, TX_LOAD,
RX_LOAD, TR) for 8 chips, then the shared SPI. 1.8 V.

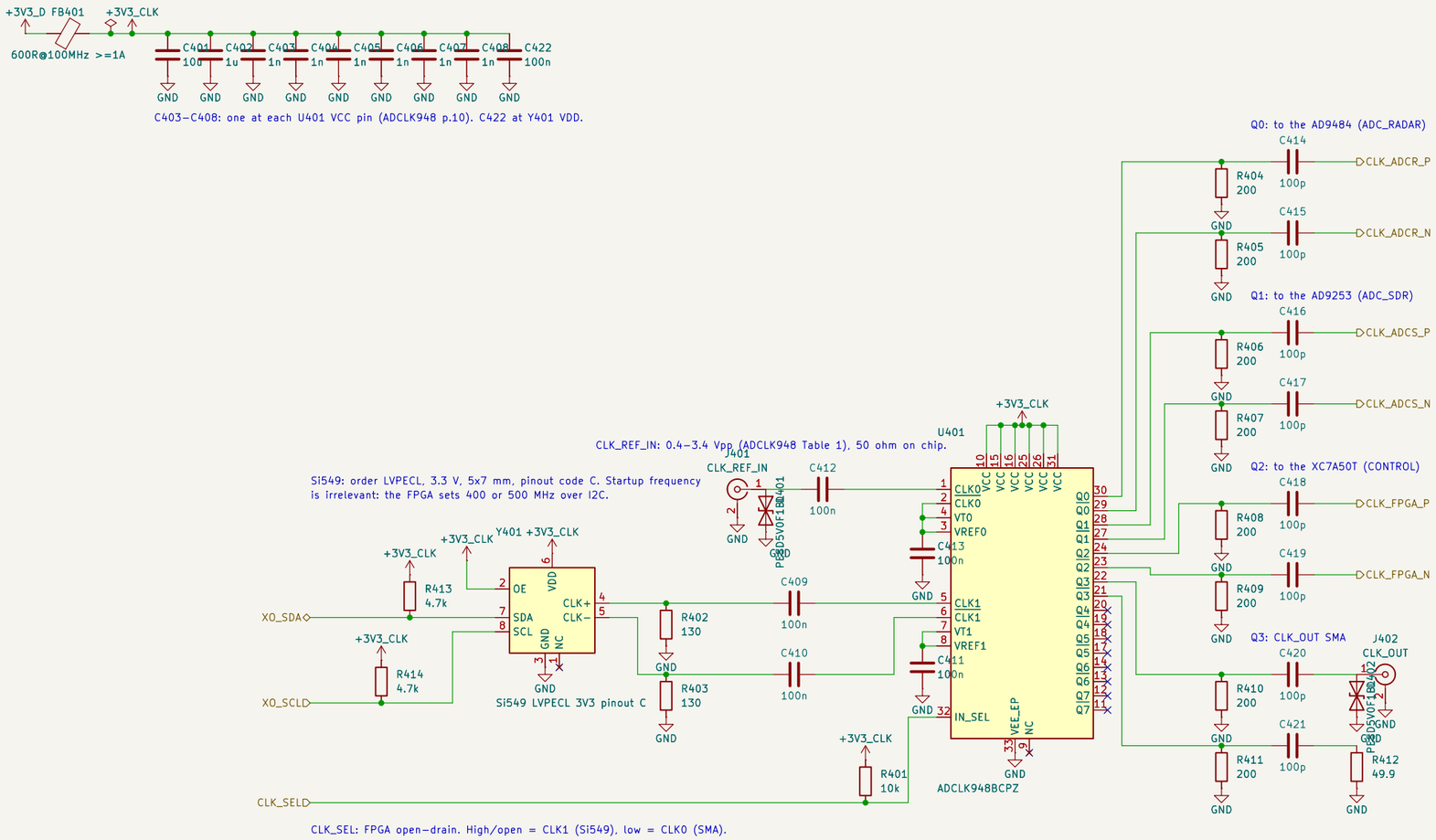
CONTROL — how this sheet works

1. Every FPGA ball has a stub and a name. Hierarchical label = the net leaves the sheet; plain label = it stays here.
2. Bank voltages: 14 and 15 at 3.3 V, 34 at 2.5 V (link outputs only), 35 at 1.8 V. See docs/CONTROL_SPEC.md.
3. Pin table: _work/sheets/control_spec.py, checked against the symbol's ball names (bank, pair, clock capable).
4. Boot: master SPI x4 from U1102 (UG470 Figure 2-14). Reset: SW1101 or the Kria's FPGA_PROG_B via U1103.

Generated by _work/sheets/control.py — review before editing by hand
KV260-IAS2RF
Sheet: /CONTROL/
File: CONTROL.kicad_sch

Title: CONTROL — XC7A50T front-end FPGA, configuration, Pmod, AERIS ribbo

Size: A1 | Date: | Rev: 03
KiCad E.D.A. 10.0.6 | Id: 3/10



CLOCK — how this sheet works

1. One master clock, 400 or 500 MHz, from the Si549 (CLK1, default) or from AERIS-10 via J401 (CLK0).
2. U401 copies it to the AD9484, the AD9253 and the FPGA. The AD9253 divides it internally (1..8).
3. The DAC clocks are NOT here: both DACs take CMOS clocks, which the FPGA generates with their data.
4. Every value on this sheet is justified in _work/sheets/clock_spec.py and docs/CLOCK_SPEC.md.

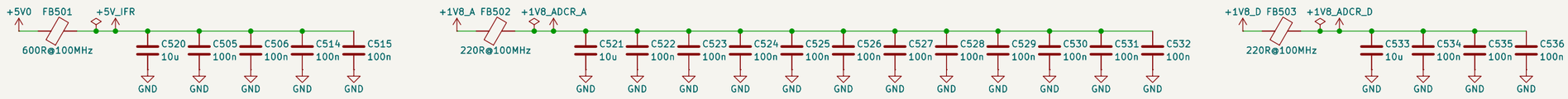
Generated by _work/sheets/clock.py — review before editing by hand

KV260-IAS2RF

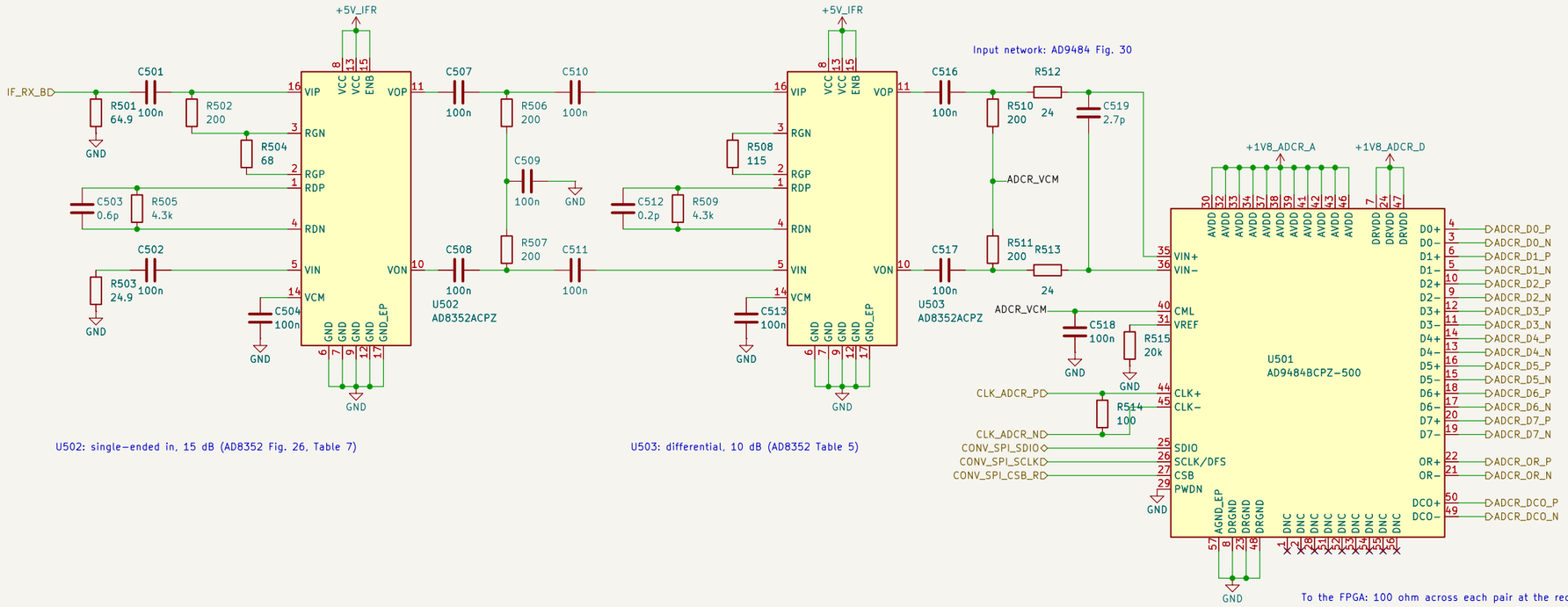
Sheet: /CLOCK/
File: CLOCK.kicad_sch

Title: CLOCK — master XO, reference select, fan-out

Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 4/10



C505/C506: at U502 VCC pins 8/13, C514/C515: at U503 VCC pins 8/13 (AD8352 eval board).
C522–C532: one at each AVDD pin, C534–C536: one at each DRVDD pin (AD9484 eval board).



U502: single-ended in, 15 dB (AD8352 Fig. 26, Table 7)

U503: differential, 10 dB (AD8352 Table 5)

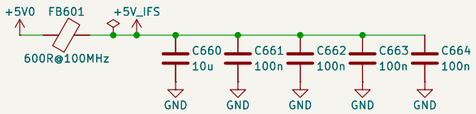
To the FPGA: 100 ohm across each pair at the receiver (AD9484 p.16).

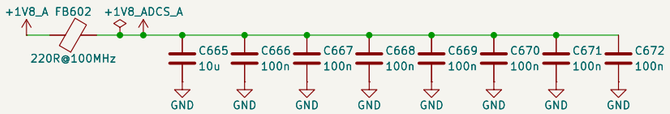
ADC_RADAR — how this sheet works

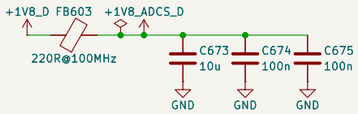
1. The radar IF (120 MHz, AERIS-10) arrives single-ended on IF_RX_B from the routing switch.
2. Two AD8352s give 25 dB, as AERIS-10's two cascaded stages; the first also converts to differential.
3. U501 samples at 400 MSPS (radar) or 500 MSPS (SDR plan): the clock comes from CLOCK, Q0.
4. CONV_SPI_* are 1.8 V signals: AD9484 abs. max DRVDD + 0.2 V. The FPGA must drive them from a 1.8 V bank.
5. Every value on this sheet is justified in _work/sheets/adc_radar_spec.py and docs/ADC_RADAR_SPEC.md.

Generated by _work/sheets/adc_radar.py — review before editing by hand
KV260-IAS2RF

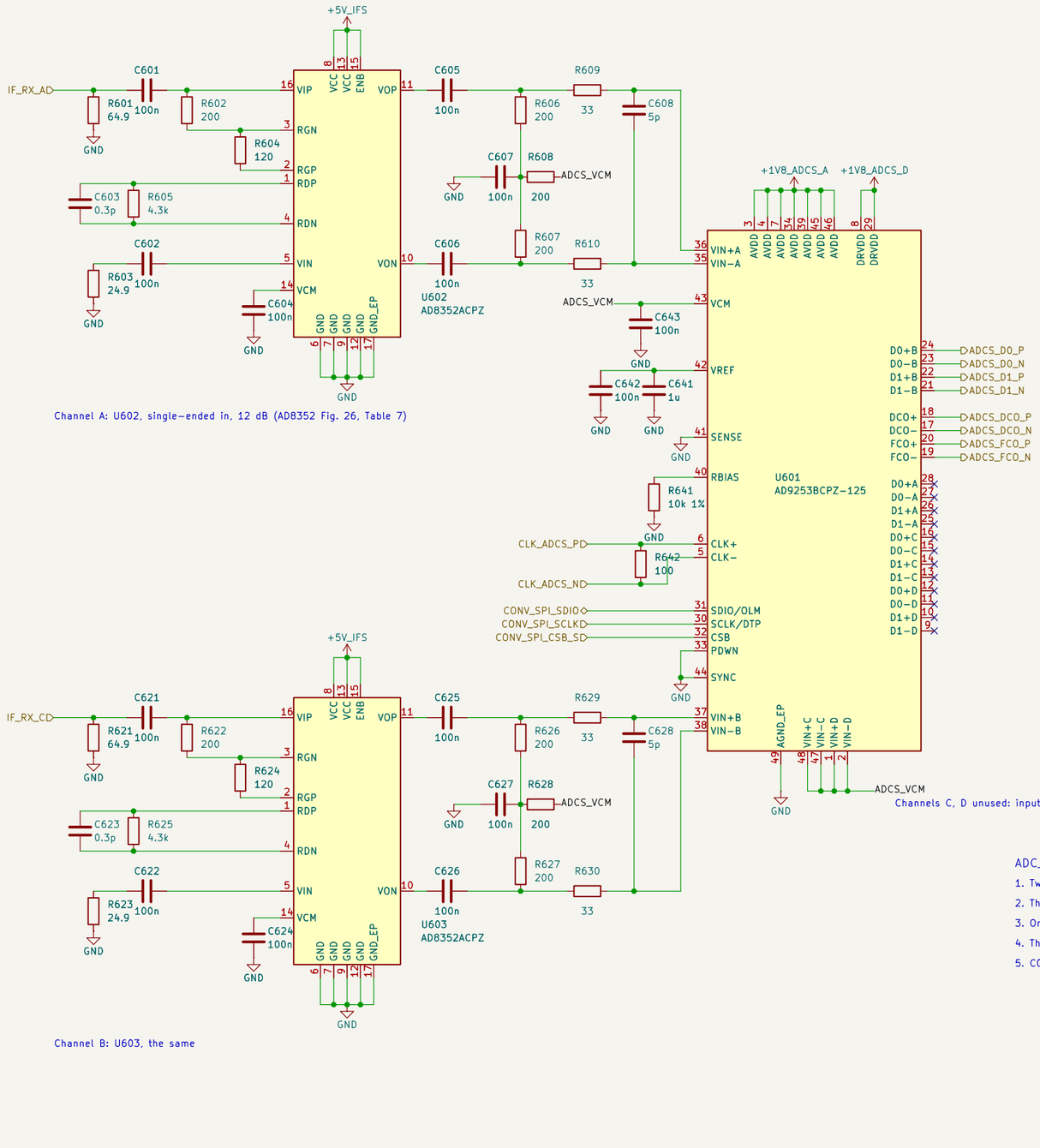
Sheet: /ADC_RADAR/		
File: ADC_RADAR.kicad_sch		
Title: ADC_RADAR — AD9484 radar ADC and its AD8352 drivers		
Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 5/10







C661/C662 at U602 VCC 8/13, C663/C664 at U603. C666–C672 one per AVDD pin, C674/C675 one per DRVDD pin.



Channel A: U602, single-ended in, 12 dB (AD8352 Fig. 26, Table 7)

Channel B: U603, the same

Channels C, D unused: inputs held at VCM

ADC_SDR — how this sheet works

1. Two SDR inputs, IF_RX_A and IF_RX_C, single-ended 50 ohm, each through an AD8352 (12 dB).
2. The AD9253 divides the 500 MHz master clock by 8: 62.5 MSPS, one-lane mode (register 0x21, SPI).
3. One-lane: channel A leaves on D0+−B, channel B on D1+−B (AD9253 Table 8, footnote 1).
4. The inputs are not biased inside the AD9253: centre taps go to its VCM pin through 200 ohm (Fig. 58).
5. CONV_SPI_* are 1.8 V signals (abs. max 2.0 V). LVDS terminated at the FPGA, 100 ohm per pair.

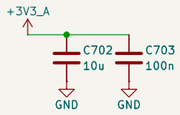
Generated by _work/sheets/adc_sdr.py — review before editing by hand

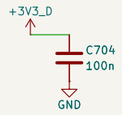
KV260-IAS2RF

Sheet: /ADC_SDR/
File: ADC_SDR.kicad_sch

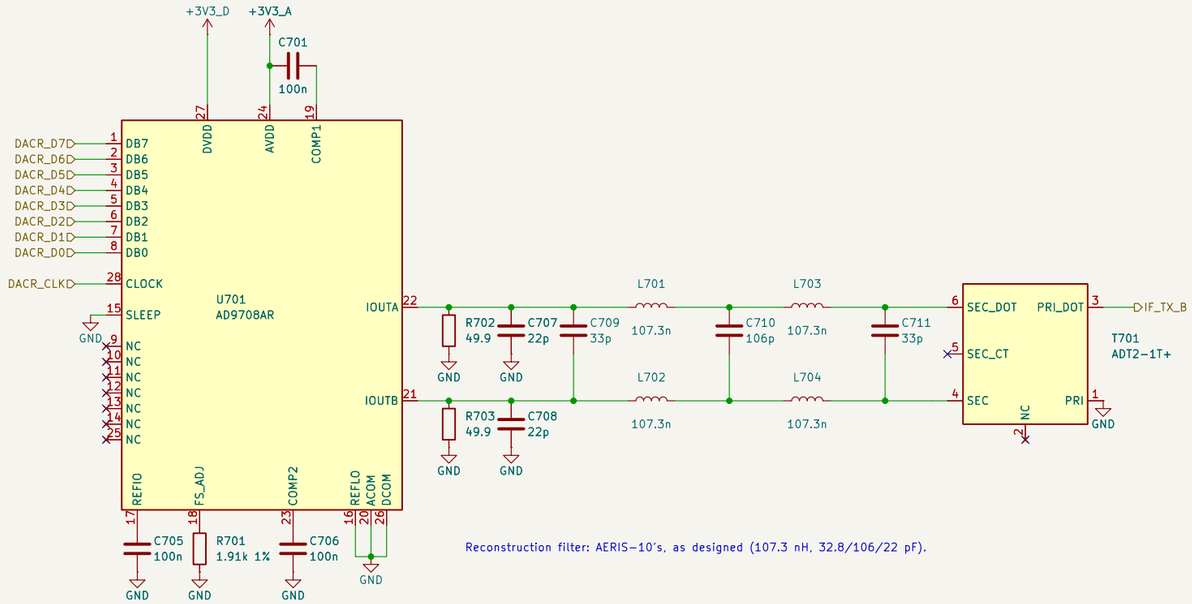
Title: ADC_SDR — AD9253 SDR ADC, two channels, AD8352 drivers

Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 6/10





+3V3_A is the analogue LDO; no extra filter (AD9708 Figure 22 is for a shared supply).



Reconstruction filter: AERIS-10's, as designed (107.3 nH, 32.8/106/22 pF).

DAC_RADAR — how this sheet works

1. The FPGA sends 8-bit samples and their clock (3.3 V CMOS); series 22 ohm sit at the FPGA end, on CONTROL.
2. Internal 1.2 V reference (REFLO to ACOM), RSET 1.91 kohm: 20 mA full scale. AERIS-10 differs here, see the spec.
3. 50 ohm loads, then AERIS-10's differential LC reconstruction filter, then a 2:1 balun to 50 ohm single-ended.
4. Every value is justified in _work/sheets/dac_radar_spec.py and docs/DAC_RADAR_SPEC.md.

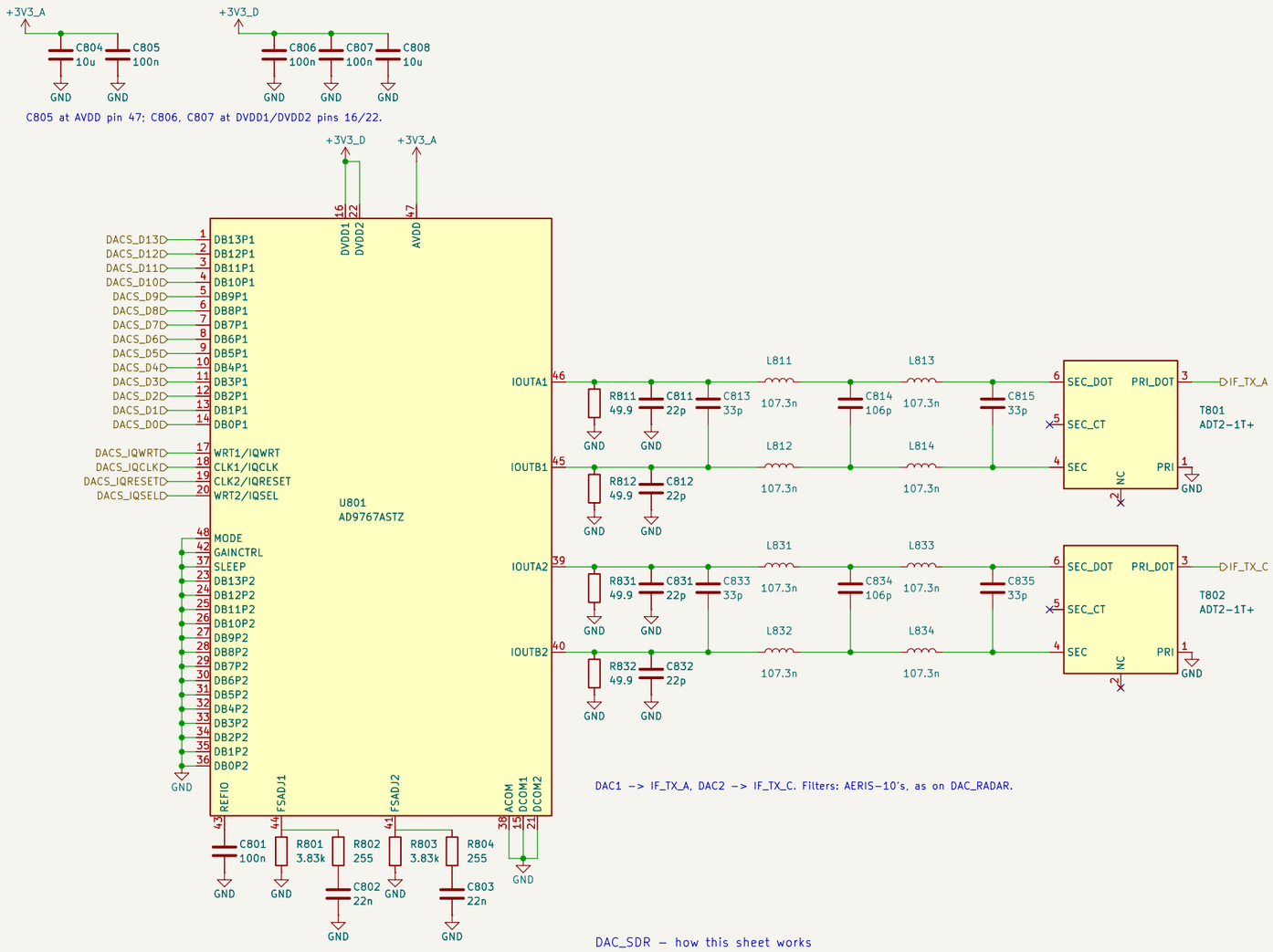
Generated by _work/sheets/dac_radar.py — review before editing by hand

KV260-IAS2RF

Sheet: /DAC_RADAR/
File: DAC_RADAR.kicad_sch

Title: DAC_RADAR — AD9708 radar chirp DAC, reconstruction filter, balun

Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 7/10



DAC1 -> IF_TX_A, DAC2 -> IF_TX_C. Filters: AERIS-10's, as on DAC_RADAR.

DAC_SDR — how this sheet works

1. Interleaved mode (MODE low): the FPGA writes I and Q alternately on one 14-bit port; IQSEL steers each word.
2. IQCLK is divided by 2 inside; IQRESET aligns the pair. Series 22 ohm on the data sit at the FPGA, on CONTROL.
3. Internal 1.2 V reference; RSET 3.83 kohm per DAC = 10 mA, 0.5 V on 49.9 ohm: best distortion (p.24).
4. Every value is justified in _work/sheets/dac_sdr_spec.py and docs/DAC_SDR_SPEC.md.

Generated by _work/sheets/dac_sdr.py — review before editing by hand

KV260-IAS2RF

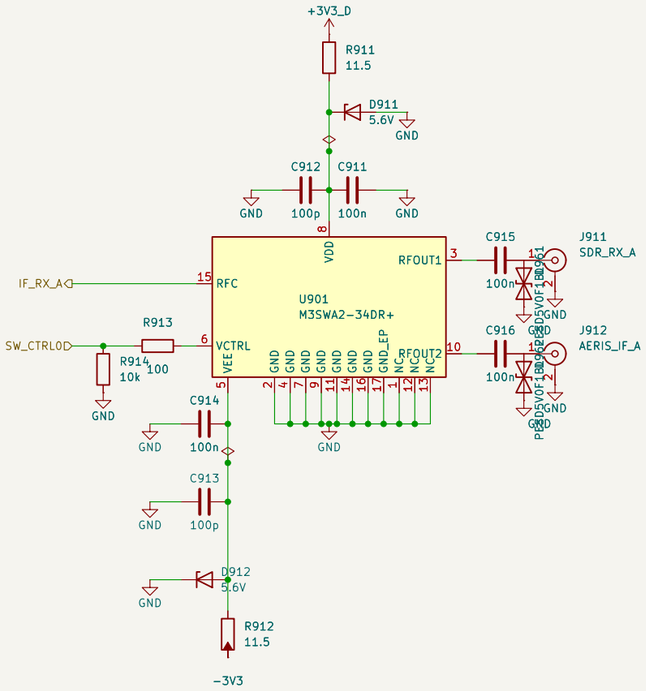
Sheet: /DAC_SDR/

File: DAC_SDR.kicad_sch

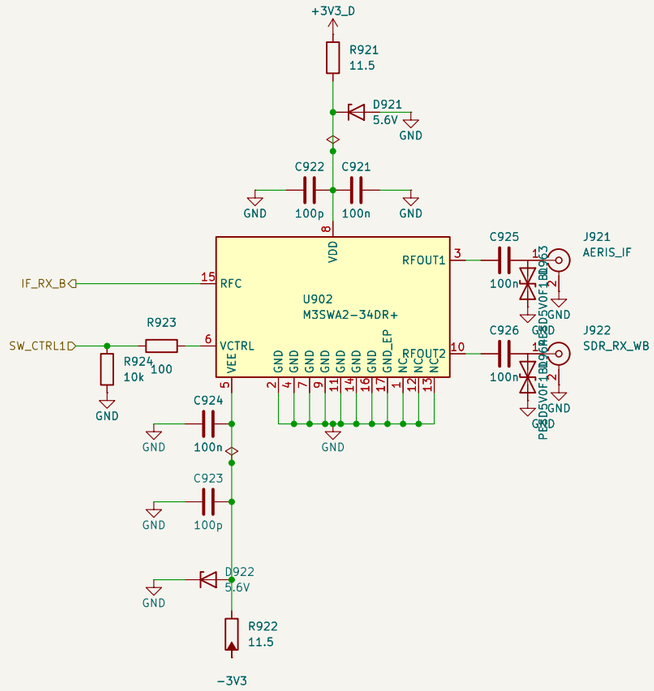
Title: DAC_SDR — AD9767 dual SDR DAC, interleaved, filters and baluns

Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 8/10

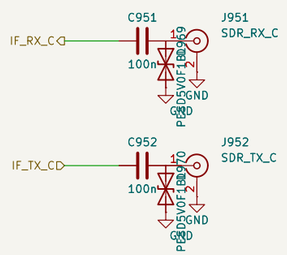
SW1: IF_RX_A. RF1 when SW_CTRL0 is low (default), RF2 when high.



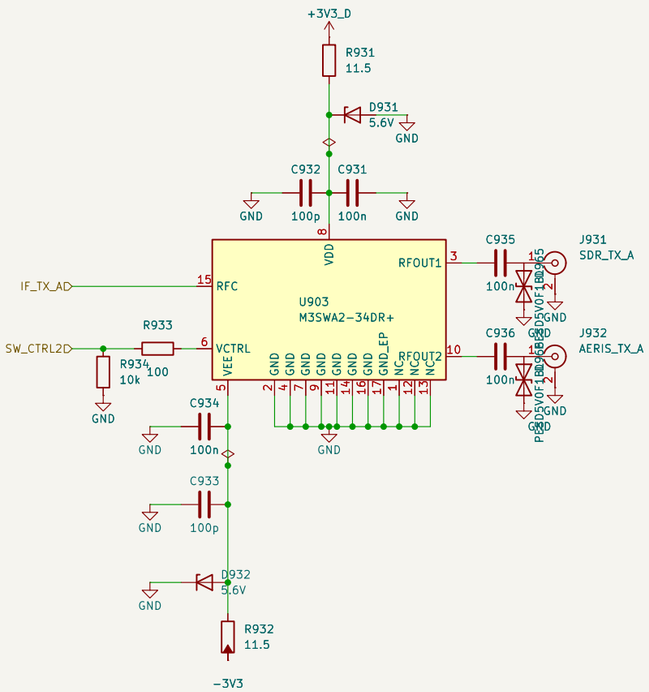
SW2: IF_RX_B. RF1 when SW_CTRL1 is low (default), RF2 when high.



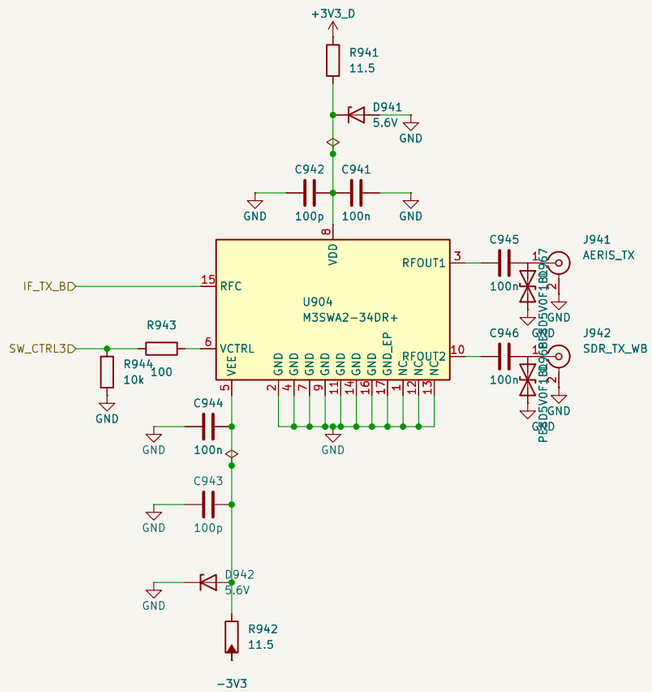
Second SDR channels: straight to their SMAs, DC-blocked.



SW3: IF_TX_A. RF1 when SW_CTRL2 is low (default), RF2 when high.



SW4: IF_TX_B. RF1 when SW_CTRL3 is low (default), RF2 when high.



ROUTING — how this sheet works

1. Each SPDT connects a converter to its own chain (RF1, default)
   or to the other one (RF2), under SW_CTRL0..3 from the FPGA.
2. Supply and control networks: M3SWA2 datasheet Figure 2, per switch.
3. Every SMA is DC-blocked: 'All RF-ports must be DC blocked or held at 0 V DC'.
4. The switches need VDD +3.3 V and VEE −3.3 V (B-03).
5. Values and sources: _work/sheets/routing_spec.py, docs/ROUTING_SPEC.md.

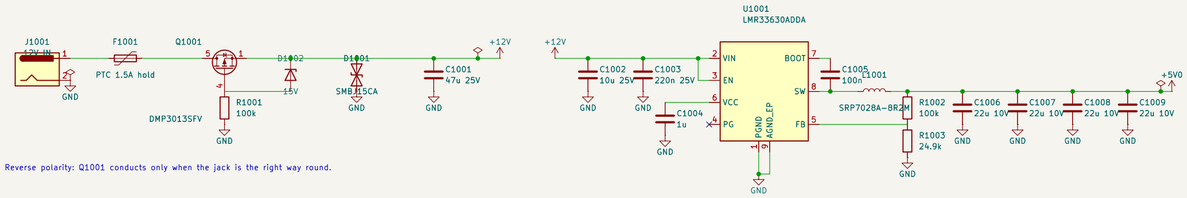
Generated by _work/sheets/routing.py — review before editing by hand

KV260-IAS2RF

Sheet: /ROUTING/
File: ROUTING.kicad_sch

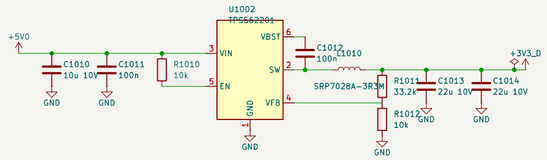
Title: ROUTING — 4 x M3SWA2 SPDT channel routing, IF SMAs

Size: A3	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 9/10

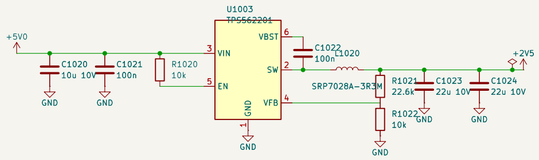


Reverse polarity: Q1001 conducts only when the jack is the right way round.

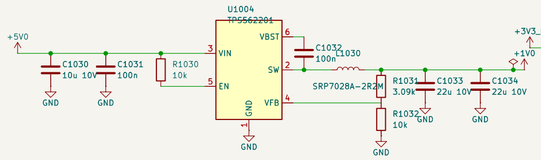
U1001 LMR33630: 12 V -> 5.0 V, 3 A, 400 kHz. Values: datasheet Table 9-2, 5 V row.



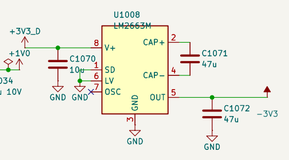
U1002 TPS562201: +5V0 -> +3V3_D, 2 A (Figure 7-1, Table 7-2).



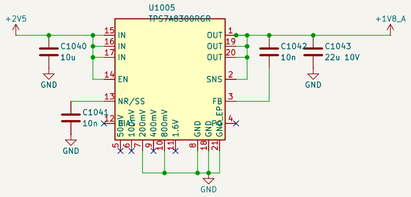
U1003 TPS562201: +5V0 -> +2V5, 2 A (Figure 7-1, Table 7-2).



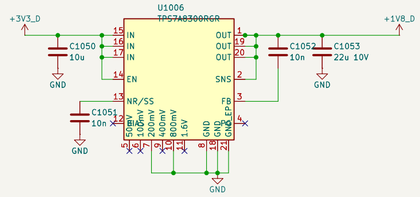
U1004 TPS562201: +5V0 -> +1V0, 2 A (Figure 7-1, Table 7-2).



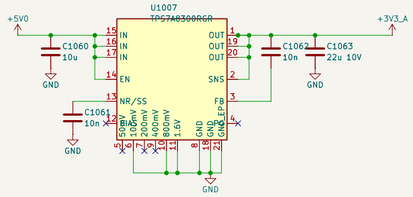
U1008 LM2663: +3V3_D -> about -3.25 V, for the RF switches' VEE.



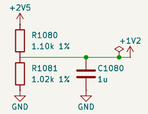
U1005 TPS7A8300: +2V5 -> +1V8_A = 0.8 + 0.2 + 0.8 V (ANY-OUT, Table 2).



U1006 TPS7A8300: +3V3_D -> +1V8_D = 0.8 + 0.2 + 0.8 V (ANY-OUT, Table 2).



U1007 TPS7A8300: +5V0 -> +3V3_A = 0.8 + 0.1 + 0.8 + 1.6 V (ANY-OUT, Table 2).



+1V2: bias for the two terminated link pairs (IAS_LINK). No DC load.

POWER — how this sheet works

1. 12 V in through a polyfuse, reverse-polarity P-MOSFET and TVS; LMR33630 makes +5V0 (3 A).
2. From +5V0: three TPS562201 bucks (+3V3_D, +2V5, +1V0) and the +3V3_A LDO.
3. TPS7A8300 LDOs: +1V8_A from +2V5, +1V8_D from +3V3_D. −3V3 from an LM2663 inverter.
4. +2V5 also powers the FPGA's LVDS banks (IAS_LINK_SPEC 3). +1V2 is a divider: bias only.
5. Every value comes from the datasheet's typical application: see docs/POWER_SPEC.md.

Generated by _work/sheets/power.py — review before editing by hand

KV260-IAS2RF

Sheet: /POWER/
File: POWER.kicad_sch

Title: POWER — 12 V input, protection, bucks, LDOs, −3V3

Size: A2	Date:	**Rev: 03**
KiCad E.D.A. 10.0.6		Id: 10/10